HW10

December 4, 2025

Exercise 1 Let $H : \mathbb{R}^n \to \mathbb{R}$ be convex with

$$L(q) := H^*(q) = \sup_{p \in \mathbb{R}^n} \{p \cdot q - H(p)\},$$

where $p \cdot q = p_1 q_1 + \cdots + p_n q_n$ is the dot product in $\mathbb{R}^n$.

1. Show that if $H(p) = \frac{1}{r}|p|^r$, $r \in (1, \infty)$, then $L(q) = \frac{1}{s}|q|^s$, with $r^{-1} + s^{-1} = 1$.
2. Let
$$H(p) = \frac{1}{2} \sum_{i,j=1}^{n} a_{ij} p_i p_j + \sum_{i=1}^{n} b_i p_i,$$
where $A = (a_{ij})$ is symmetric, positive definite. Compute $L(q)$.

Exercise 2 Let $H : \mathbb{R} \to \mathbb{R}$ be convex and $L = H^*$. We say $q \in \partial H(p)$, if

$$H(r) \geq H(p) + q \cdot (r - p), \quad \forall r \in \mathbb{R}.$$

Show that $q \in \partial H(p)$ if and only if $p \in \partial L(q)$, if and only if $p \cdot q = H(p) + L(q)$.

Exercise 3 Let E be a closed subset of $\mathbb{R}^n$. Apply the Hopf–Lax formula to solve the initial value problem

$$\begin{cases} u_t + |\nabla u|^2 = 0, & (t, x) \in (0, \infty) \times \mathbb{R}^n, \\ u = 0, \ x \in E, \quad u = +\infty, \ x \notin E, & t = 0. \end{cases}$$

and show that the solution is $u(t, x) = \frac{1}{4t} \operatorname{dist}(x, E)^2$.

Exercise 4 Use the Hopf–Lax formula to show that, if u^i, $i = 1, 2$, solve

$$\begin{cases} u_t^i + H(\partial_x u^i) = 0, & (0, \infty) \times \mathbb{R}, \\ u^i = g^i, & \{t = 0\} \times \mathbb{R}, \end{cases}$$

then the L^∞-contraction holds

$$\sup_{x \in \mathbb{R}} |u^1(t, x) - u^2(t, x)| \leq \sup_{x \in \mathbb{R}} |g^1(x) - g^2(x)|, \quad t > 0.$$

Ex1. (1) 由 Young 不等式，$p\cdot q - H(p) \le |p||q| - \frac{1}{r}|p|^r \le \frac{1}{s}|q|^s$，故 $L(q) \le \frac{1}{s}|q|^s$

注意到 $\exists p, q$ 使等号成立，故上确界可达，即 $L(q) = \frac{1}{s}|q|^s$

(2) 令 $f(p) = p^T q - H(p)$，其关于 p 下凸 (convex)

则 $\nabla_p f = q - Ap - b = 0 \Rightarrow$ 最值在 $p^* = A^{-1}(q-b)$ 处达到

从而 $L(q) = (p^*)^T q - \frac{1}{2}(p^*)^T A p^* - b^T p^*$

$$= (p^*)^T(q-b) - \frac{1}{2}(p^*)^T A p^*$$

$$= \frac{1}{2}(p^*)^T A p^*$$

$$= \frac{1}{2}(q-b)^T A^{-1}(q-b)$$

Ex2. 一方面，若 $q \in \partial H(p)$，则 $H(r) \ge H(p) + q\cdot(r-p)$，$\forall r \in \mathbb{R}$

即 $p\cdot q - H(p) \ge r\cdot q - H(r)$，$\forall r \in \mathbb{R}$，即 $L(q) \le p\cdot q - H(p)$.

又由 $L(q)$ 的定义知，$L(q) \ge p\cdot q - H(p)$

移项即得 $q \in \partial H(p) \Rightarrow p\cdot q = H(p) + L(q)$

另一方面，若 $p\cdot q = H(p) + L(q)$，由 $L(q)$ 的定义可得 $H(r) \ge H(p) + q(r-p)$，$\forall r \in \mathbb{R}$

即 $p\cdot q = H(p) + L(q) \Rightarrow q \in \partial H(p)$

故得 $q \in \partial H(p) \Leftrightarrow p\cdot q = H(p) + L(q)$

又由 $H = H^{**} = L^*$ 知，可类似得 $p \in \partial L(q) \Leftrightarrow p\cdot q = H(p) + L(q)$，从而结论得证.

Ex3. 取 $H(p) = |p|^2$，则 $L(q) = \frac{1}{4}|q|^2$

从而由 Hopf-Lax formula 得 $u(t,x) = \frac{1}{4t}\min_{y\in E}|x-y|^2 = \frac{1}{4t}\,\mathrm{dis}(x,E)^2$

Ex4. 由 Hopf-Lax formula 知，$u_i(t,x) = \inf_{y\in\mathbb{R}}\{g_i(y) + tL(\frac{x-y}{t})\}$，$i=1,2$

记 $G := \sup_{x\in\mathbb{R}}|g_1(x) - g_2(x)|$，从而 $g_i(x) \le g_j(x) + G$ $\quad i\ne j$ 且 $i,j \in \{1,2\}$，$x\in\mathbb{R}$

故 $g_1(y) + tL(\frac{x-y}{t}) \le g_2(y) + tL(\frac{x-y}{t}) + G$

从而 $u_1(t,x) = \inf_{y\in\mathbb{R}}\{g_1(y) + tL(\frac{x-y}{t})\} \le \inf_{y\in\mathbb{R}}\{g_2(y) + tL(\frac{x-y}{t})\} + G = u_2(t,x) + G$

类似有 $u_2(t,x) \le u_1(t,x) + G$

从而 $|u_1(t,x) - u_2(t,x)| \le G$，两边关于 x 取 sup 即得结论.